Predicting 30-Day Hospital Readmission Using Synthetic EHR Data

Building a machine learning pipeline to identify high-risk patients and reduce costly hospital readmissions

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The Clinical Challenge

Financial Impact

Hospital readmissions cost healthcare systems billions annually. CMS imposes penalties on facilities with excessive 30-day readmission rates.

Quality Metric

Readmission rates serve as a key indicator of care quality and patient outcomes.

Our Goal

Develop an ML pipeline to flag high-risk patients for targeted interventions.

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Dataset Overview

1,171

Patients

Unique individuals in the synthetic EHR

3,928

Eligible Discharges

Discharge records meeting study criteria

20%

Readmission Rate

Patients readmitted within 30 days

Source: Synthea synthetic EHR data — realistic but privacy-protected records generated using clinical rules and statistical models

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Existing Literature

Two reference studies using similar methods on real EHR data provide benchmark performance expectations:

Study 1

Similar methodology applied to real EHR data

AUC: 0.667

Study 2

Comparative approach with different feature sets

AUC: 0.757

These benchmarks help contextualize our model performance expectations.

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Technical Stack

Data Query

SQL for database extraction

Data Processing

Polars for efficient transformations

ML Framework

scikit-learn for model development

Tree Models

XGBoost for gradient boosting

Deep Learning

TensorFlow/Keras for neural networks

Visualization

matplotlib for plotting results

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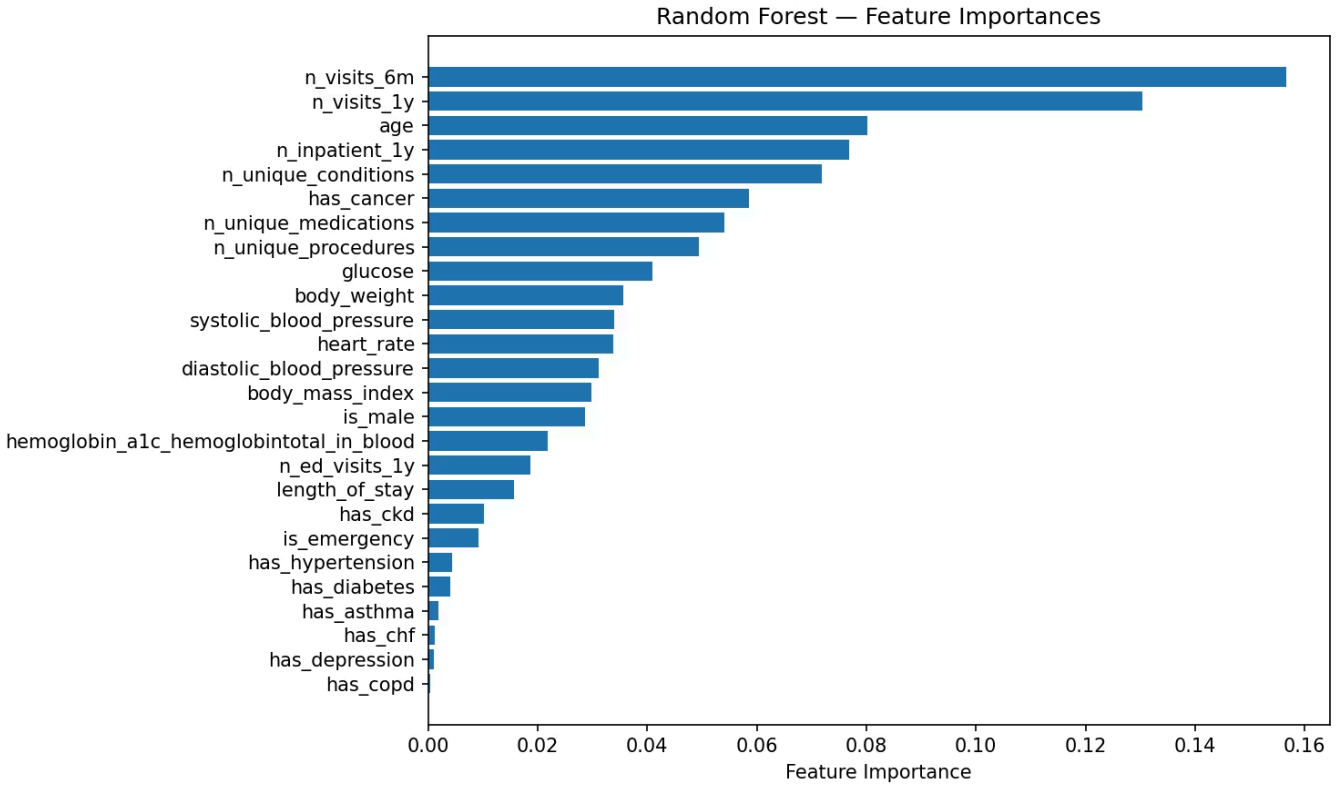
Feature Engineering Strategy

26 Features Across 6 Categories

- Demographics
- Clinical history
- Laboratory values
- Medications
- Vital signs
- Procedures

Key Constraint

All features use only **pre-admission data** to prevent data leakage



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Technical Challenges Overcome

Data Leakage

Ensured temporal consistency by using only pre-admission data

Case Sensitivity

Fixed keyword matching bugs in SQL queries

Multicollinearity

Identified and addressed correlated features

Patient-Level Split

Split by patient ID to prevent data leakage

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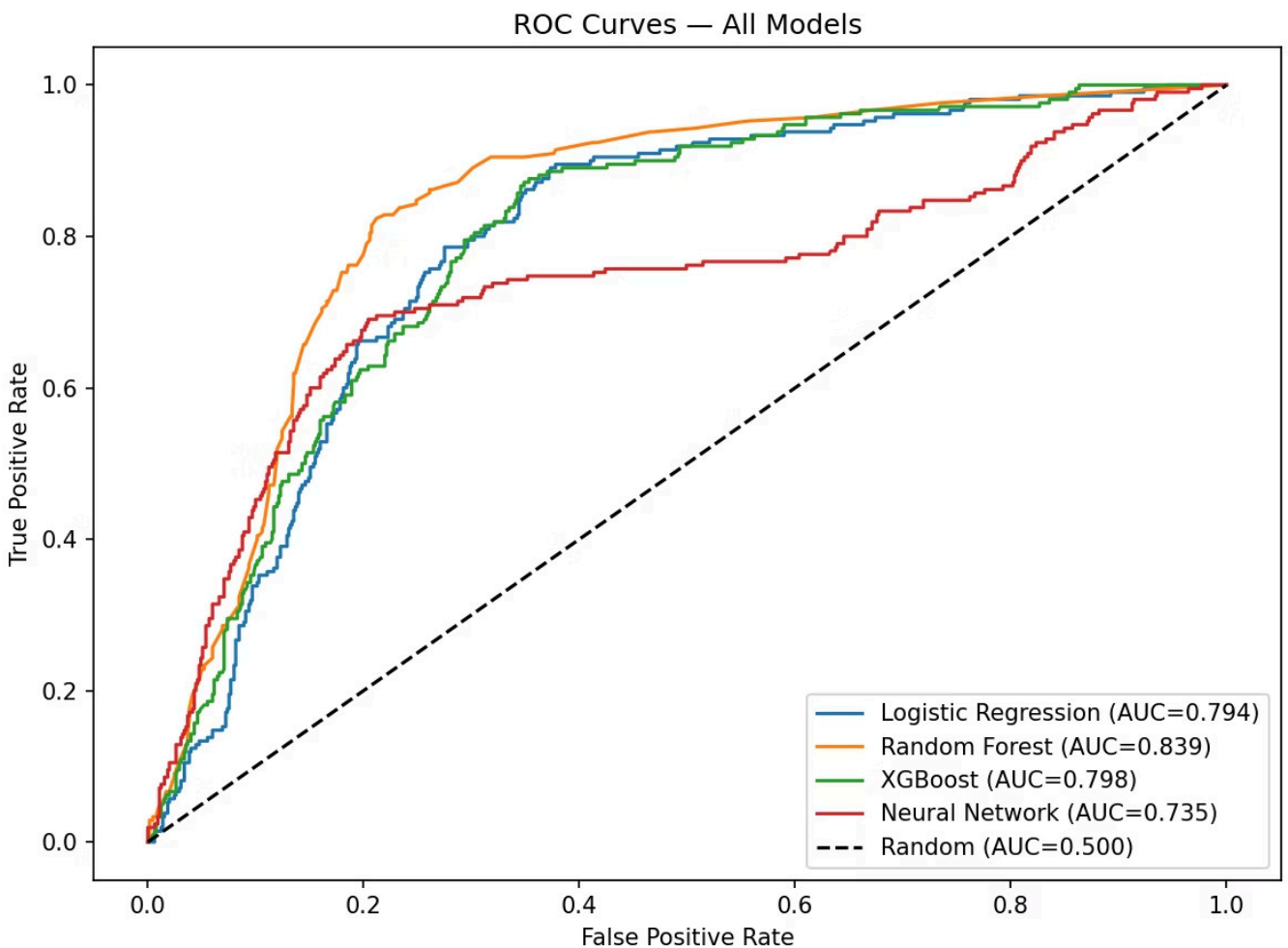
Class Imbalance & Model Calibration

Addressing Imbalance

- Class weighting in models
- Stratified sampling
- Threshold tuning

Calibration vs Discrimination

Balanced calibration (reliable probabilities) against discrimination (ranking ability)



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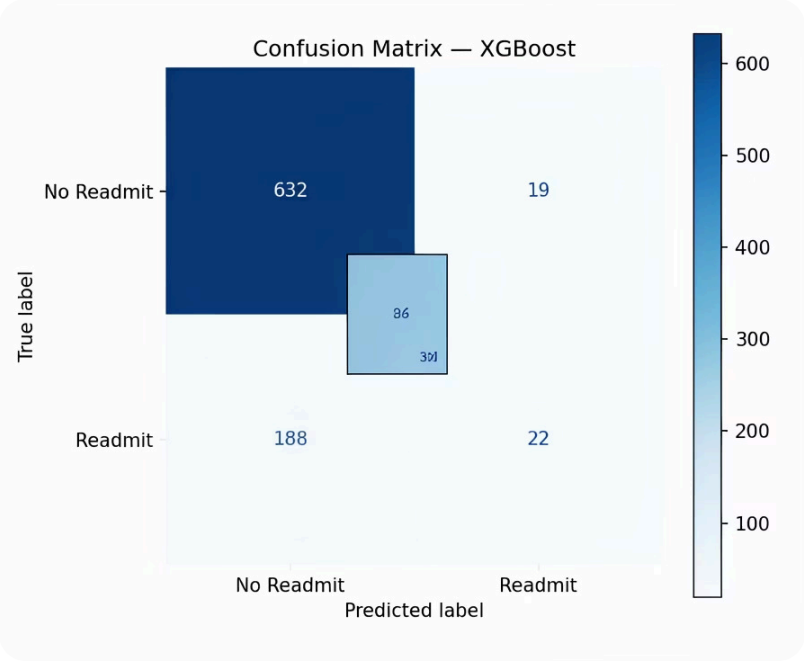
Results Summary

Model Performance

Model	CV AUC	Test AUC	Recall	Precision
LogReg	0.873	0.794	55%	51%
RandForest	0.858	0.839	7%	52%
XGBoost	0.876	0.798	10%	54%
NN	N/A	0.735	57%	56%



XGBoost achieved the highest CV AUC (0.876), Random Forest the highest test AUC (0.839), and the neural network the best recall (57%). No single model dominated all metrics.

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Limitations & Future Directions

1. **Current Limitation**

   Synthetic data (Synthea) lacks real-world complexity

2. **Next Step**

   Validate pipeline on real EHR data (MIMIC-IV)

3. **Future Work**

   External validation, prospective clinical trials

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